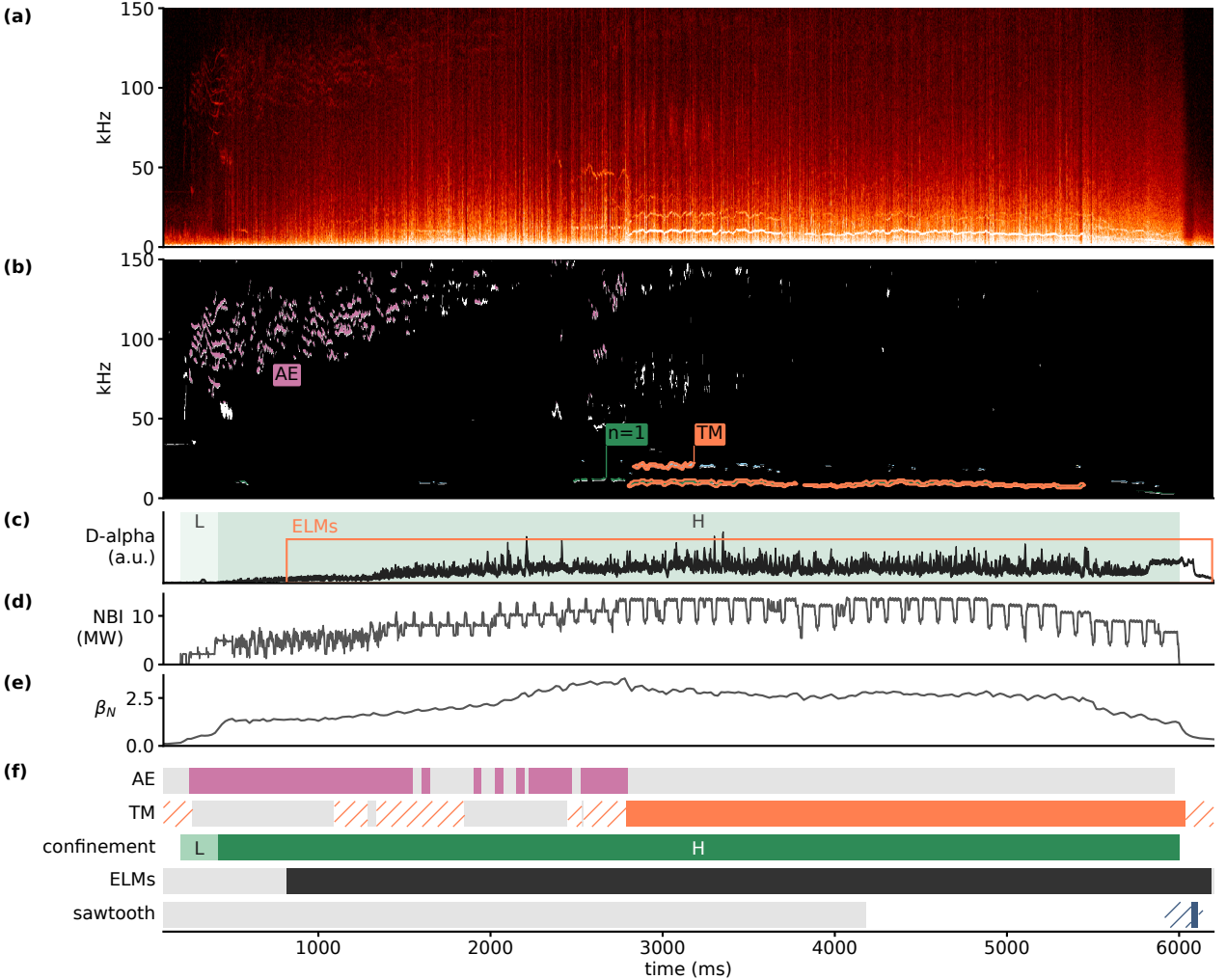

(a)
150
100
50
0
kHz
(b)
150
100
50
0
kHz
AE
n=1
TM
(c)
D-alpha
(a.u.)
L
ELMs
H
(d)
NBI
(MW)
10
0
(e)
$\beta_N$
2.5
0.0
(f)
AE
TM
confinement
L
H
ELMs
sawtooth
1000
2000
3000
4000
5000
6000
time (ms)